

Introduction

Basics of DC versus AC power and resistive versus reactive loads.

What is a Phasor?

Relationship between sinusoids and phasors. Deriving the concept of phasors. Formal phasor transformation and notation.

Life With and Without Phasors: Example Circuit Analysis

Seeing the benefits of phasors in practice over differential equations and the Laplace transform.

Tutorial: Building an AC Circuit Analyzer

Developing AC circuit analysis skills by building a software tool to solve any AC circuit.

Complex Power and the Power Factor

Deriving complex power and the power triangle from time-domain instantaneous power. Power factor and power factor correction.

Three-Phase Power

Motivation for three-phase power. Wye versus delta configurations. Three-phase power in the grid.

Grid Modeling

Bus injection model. Power flow, optimal power flow, economic dispatch, and unit commitment problems.